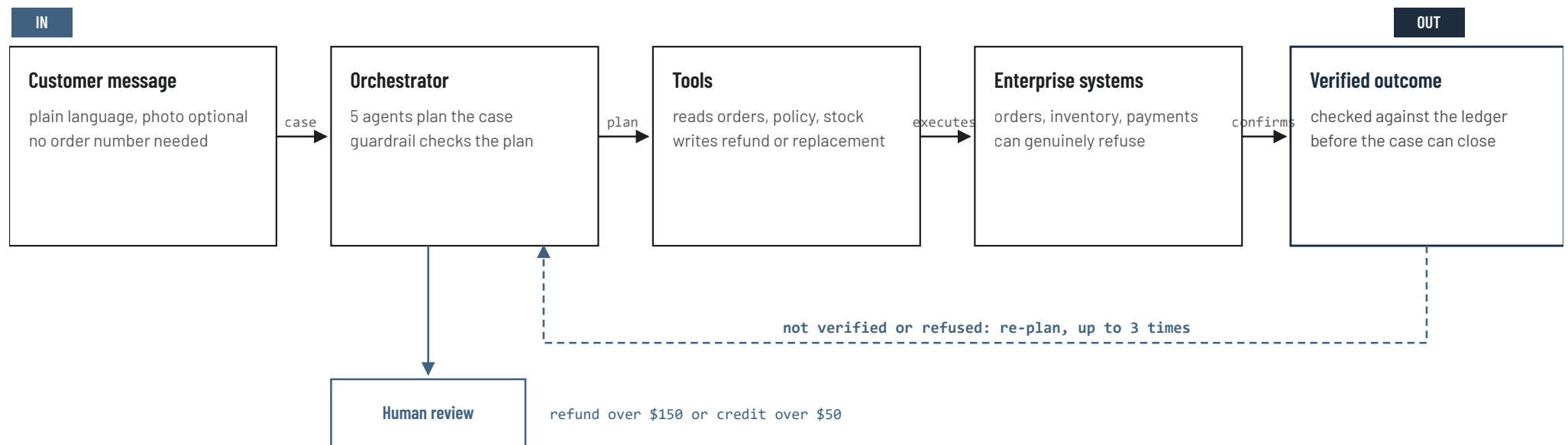


# One case, end to end

A customer message enters on the left with no order number. A verified state change leaves on the right, or a human is asked, or the agent refuses. Every box below is code in the repository, not an intention.



**Two databases, deliberately separate.** desk.db is the ticketing product; enterprise.db is the company the agent acts on. The agent reaches the second only through tools, and the verifier reads it directly, so no agent can close a case by claiming success.

## Where each required element lives

| ELEMENT            | COMPONENT                         | HOW IT WORKS                                                                                                                                                                                          |
|--------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent / controller | agents/orchestrator.py            | Explicit state machine with a bounded re-plan loop, an approval gate and one daemon thread per case.                                                                                                  |
| Tools              | tools/read_tools.py · actions.py  | Read-only tools scoped per agent and executed concurrently within a round; five typed write actions only the Executor may call.                                                                       |
| External systems   | sandbox/services.py               | CRM, orders, per-warehouse inventory with transit times, payment gateway, carrier, store credit, each able to genuinely refuse.                                                                       |
| Memory / state     | db.py · sandbox/store.py          | Case state and a complete agent trace; a learned knowledge-base entry per resolved case; an independent enterprise audit log.                                                                         |
| Retrieval          | search_policy · search_past_cases | Scored keyword retrieval over the policy handbook and previously resolved cases. The Communicator writes back, so the next case starts better informed.                                               |
| Planning           | Resolver Agent                    | Typed multi-step plans carrying an explicit expected outcome; validated, then independently audited before anything runs.                                                                             |
| Evaluation         | sandbox/verify.py                 | SQL checks against the systems of record, including a no_over_refund invariant on every case. A failed check re-enters planning.                                                                      |
| Human interaction  | Portal · desk · proposals         | Customers file and attach evidence; supervisors approve or reject anything past the guardrail, with the reason and exact actions shown.                                                               |
| Failure handling   | orchestrator.py · llm.py          | Tool errors returned as data; retryable errors retried on the same idempotency key; a guard against repeating a refused action; provider failover with backoff; deterministic offline agents beneath. |

# The loop that makes it agentic

|                                                                                                                                                                                                    |                                                                                                                                                                   |                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>Refusal is evidence</b><br>An action the enterprise rejects re-enters planning as a new fact. A plan repeating an action already refused with identical parameters is blocked before execution. | <b>Done is proven</b><br>A case closes only when SQL against the systems of record matches the plan's promised outcome. The agent does not get to assert success. | <b>Limits are code</b><br>Approval thresholds live in Python, never in a prompt, so no message from a customer can widen them. |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|

Bounded by MAX\_AGENT\_STEPS, MAX\_AUDIT\_ROUNDS = 2 and MAX\_REPLANS = 3; when the budget is spent the case escalates carrying everything it learned, so a human never restarts from zero.